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ROBOTIC SURGICAL SYSTEM, CONTROL METHOD FOR ROBOTIC SURGICAL SYSTEM, AND STORAGE MEDIUM

Abstract

A robotic surgical system includes a controller configured or programmed to restrict an operation to rotate a shaft in a direction out of a rotational operating range of the shaft with an operation unit and an operation to bend a distal end device with respect to the shaft with the operation unit when an operation for a surgical instrument received by the operation unit includes an operation to rotate the shaft out of the rotational operating range of the shaft.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The priority application number JP2024-030881, Robotic Surgical System and Control Method for Robotic Surgical System, Mar. 1, 2024, KOBAYASHI Ayataka, MIZOHATA Yuichi, and YAMAMOTO Daisuke, upon which this patent application is based, is hereby incorporated by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present disclosure relates to a robotic surgical system, a control method for a robotic surgical system, and a storage medium.

Description of the Background Art

[0003] Conventionally, a robotic surgical system including a surgical instrument attached to the distal end of a robot arm and an operation unit to receive an operation for the surgical instrument is known. Japanese Patent No. 6000641 discloses a master-slave manipulator including a slave manipulator arm to which a treatment tool is attached, a manipulation input device including a manipulation handle, and a controller. In Japanese Patent No. 6000641, the treatment tool is remotely controlled by the manipulation handle. In Japanese Patent No. 6000641, the treatment tool is a pair of forceps, for example. Furthermore, in Japanese Patent No. 6000641, the controller monitors a difference between a command value for opening and closing the pair of forceps from the manipulation handle and the actual opening and closing angle of the pair of forceps. When the difference between the command value for opening and closing the pair of forceps and the actual opening and closing angle of the pair of forceps exceeds an allowable value, an interlock mode in which a manipulation for the treatment tool with the manipulation handle is received transitions to an interlock stop mode in which a manipulation for the treatment tool with the manipulation handle is not received. Furthermore, in Japanese Patent No. 6000641, after the controller detects that the treatment tool has been opened to its maximum opening and closing angle by the manipulation handle in the interlock stop mode, the treatment tool is closed again with the manipulation handle, and the controller transitions again to the interlock mode based on the difference between the command value for opening and closing the pair of forceps and the actual opening and closing angle of the pair of forceps becoming equal to or smaller than the allowable value.

[0004] However, in Japanese Patent No. 6000641, the controller switches between the interlock stop mode and the interlock mode based on the difference between the command value for opening and closing the pair of forceps from the manipulation handle and the actual opening and closing angle of the pair of forceps, but Japanese Patent No. 6000641 does not mention a case in which a manipulation is performed by the manipulation handle that attempts to exceed the movable range of the treatment tool. When the interlock stop mode is set each time the manipulation handle exceeds the movable range of the treatment tool, surgery is interrupted each time and efficient surgery cannot be performed. On the other hand, when the interlock mode is maintained even when the manipulation handle exceeds the movable range of the treatment tool, the posture of the treatment tool does not change, but the posture of the manipulation handle changes. In such a case, when a manipulation received by the manipulation handle returns to a manipulation within the movable range of the treatment tool, the posture of the treatment tool changes suddenly to follow the posture of the manipulation handle. From the above, it is desired to reduce or prevent a sudden change in the posture of the treatment tool while the interlock mode is maintained, even when a manipulation is performed by the manipulation handle that exceeds the movable range of the treatment tool.

SUMMARY OF THE INVENTION

[0005] The present disclosure is intended to provide a robotic surgical system, a control method for

a robotic surgical system, and a storage medium each capable of reducing or preventing a sudden change in the posture of a surgical instrument while a state in which an operation with an operation unit can be received is maintained even when an operation is performed that attempts to exceed the movable range of the surgical instrument.

[0006] A robotic surgical system according to a first aspect of the present disclosure includes a robot arm to allow a surgical instrument including a shaft and a distal end device connected to a distal end of the shaft via a wrist joint to be attached thereto, an operation apparatus including an operation unit to receive an operation for the surgical instrument, and a controller configured or programmed to restrict an operation to rotate the shaft in a direction out of a rotational operating range of the shaft with the operation unit and an operation to bend the distal end device with respect to the shaft with the operation unit when the operation for the surgical instrument received by the operation unit includes an operation to rotate the shaft out of the rotational operating range of the shaft.

[0007] In the robotic surgical system according to the first aspect of the present disclosure, the controller is configured or programmed to restrict the operation to rotate the shaft in the direction out of the rotational operating range of the shaft with the operation unit and the operation to bend the distal end device with respect to the shaft with the operation unit when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft. Accordingly, even when an operation is performed to rotate the shaft out of the rotational operating range of the shaft, the operation to rotate the shaft in the direction out of the rotational operating range of the shaft with the operation unit and the operation to bend the distal end device with respect to the shaft with the operation unit are restricted, and thus the posture of the operation unit is maintained. Therefore, the possibility that the postures of the distal end device and the shaft do not match the posture of the operation unit is reduced or prevented, and thus a state in which the postures of the distal end device and the shaft match the posture of the operation unit is maintained. In addition, the controller only restricts the operation to rotate the shaft in the direction out of the rotational operating range with the operation unit and the operation to bend the distal end device with respect to the shaft with the operation unit, but does not restrict reception of all operations with the operation unit. Therefore, even when an operation is performed that attempts to exceed the movable range of the surgical instrument, a sudden change in the posture of the surgical instrument can be reduced or prevented while a state in which an operation with the operation unit can be received is maintained.

[0008] A control method for a robotic surgical system according to a second aspect of the present disclosure is a control method for a robotic surgical system including a robot arm to allow a surgical instrument including a shaft and a distal end device connected to a distal end of the shaft via a wrist joint to be attached thereto, and an operation apparatus including an operation unit to receive an operation for the surgical instrument, and includes receiving the operation for the surgical instrument by the operation unit, and restricting an operation to rotate the shaft in a direction out of a rotational operating range of the shaft with the operation unit and an operation to bend the distal end device with respect to the shaft with the operation unit when the operation for the surgical instrument received by the operation unit includes an operation to rotate the shaft out of the rotational operating range of the shaft.

[0009] As described above, the control method for the robotic surgical system according to the second aspect of the present disclosure includes restricting the operation to rotate the shaft in the direction out of the rotational operating range of the shaft with the operation unit and the operation to bend the distal end device with respect to the shaft with the operation unit when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft. Accordingly, even when an operation is performed to rotate the shaft out of the rotational operating range of the shaft, the operation to rotate the shaft in the direction out of the rotational operating range of the shaft with the operation unit and the

operation to bend the distal end device with respect to the shaft with the operation unit are restricted, and thus the posture of the operation unit is maintained. Therefore, the possibility that the postures of the distal end device and the shaft do not match the posture of the operation unit is reduced or prevented, and thus a state in which the postures of the distal end device and the shaft match the posture of the operation unit is maintained. In addition, the controller only restricts the operation to rotate the shaft in the direction out of the rotational operating range with the operation unit and the operation to bend the distal end device with respect to the shaft with the operation unit, but does not restrict reception of all operations with the operation unit. Therefore, it is possible to provide the control method for the robotic surgical system capable of reducing or preventing a sudden change in the posture of the surgical instrument while a state in which an operation with the operation unit can be received is maintained even when an operation is performed that attempts to exceed the movable range of the surgical instrument.

[0010] A storage medium according to a third aspect of the present disclosure is a storage medium operable to store a program for a control method for a robotic surgical system including a robot arm to allow a surgical instrument including a shaft and a distal end device connected to a distal end of the shaft via a wrist joint to be attached thereto, and an operation apparatus including an operation unit to receive an operation for the surgical instrument, and is operable to store the program for the control method including receiving the operation for the surgical instrument by the operation unit, and restricting an operation to rotate the shaft in a direction out of a rotational operating range of the shaft with the operation unit and an operation to bend the distal end device with respect to the shaft with the operation unit when the operation for the surgical instrument received by the operation unit includes an operation to rotate the shaft out of the rotational operating range of the shaft.

[0011] As described above, the storage medium according to the third aspect of the present disclosure is operable to store the program for the control method including restricting the operation to rotate the shaft in the direction out of the rotational operating range of the shaft with the operation unit and the operation to bend the distal end device with respect to the shaft with the operation unit when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft. Accordingly, even when an operation is performed to rotate the shaft out of the rotational operating range of the shaft, the operation to rotate the shaft in the direction out of the rotational operating range of the shaft with the operation unit and the operation to bend the distal end device with respect to the shaft with the operation unit are restricted, and thus the posture of the operation unit is maintained. Therefore, the possibility that the postures of the distal end device and the shaft do not match the posture of the operation unit is reduced or prevented, and thus a state in which the postures of the distal end device and the shaft match the posture of the operation unit is maintained. In addition, only the operation to rotate the shaft in the direction out of the rotational operating range with the operation unit and the operation to bend the distal end device with respect to the shaft with the operation unit are restricted, but reception of all operations with the operation unit is not restricted. Therefore, it is possible to provide the storage medium capable of reducing or preventing a sudden change in the posture of the surgical instrument while a state in which an operation with the operation unit can be received is maintained even when an operation is performed that attempts to exceed the movable range of the surgical instrument.

[0012] According to the present disclosure, even when an operation is performed that attempts to exceed the movable range of the surgical instrument, a sudden change in the posture of the surgical instrument can be reduced or prevented while a state in which an operation with the operation unit can be received is maintained.

[0013] The foregoing and other objects, features, aspects and advantages of the present disclosure will become more apparent from the following detailed description of the present disclosure when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- [0014] FIG. **1** is a diagram showing the configuration of a robotic surgical system according to an embodiment.
- [0015] FIG. **2** is a diagram showing a display of a medical cart according to the embodiment.
- [0016] FIG. **3** is a diagram showing the configuration of the medical cart according to the embodiment.
- [0017] FIG. **4** is a diagram showing the configuration of a robot arm according to the embodiment.
- [0018] FIG. **5** is a diagram showing an instrument.
- [0019] FIG. **6** is a perspective view showing the configuration of an arm operation unit according to the embodiment.
- [0020] FIG. **7** is a diagram for illustrating translational movement of the robot arm.
- [0021] FIG. **8** is a diagram for illustrating rotational movement of the robot arm.
- [0022] FIG. **9** is a diagram showing an endoscope.
- [0023] FIG. **10** is a diagram showing a pivot position setting instrument.
- [0024] FIG. **11** is a diagram showing operation units according to the embodiment.
- [0025] FIG. **12** is a diagram showing a right-handed wrist according to the embodiment.
- [0026] FIG. **13** is a diagram showing a left-handed wrist according to the embodiment.
- [0027] FIG. **14** is a perspective view showing foot pedals according to the embodiment.
- [0028] FIG. **15** is a control block diagram of the robotic surgical system according to the embodiment.
- [0029] FIG. **16** is a control block diagram of the robot arm according to the embodiment.
- [0030] FIG. **17** is a control block diagram of a positioner and the medical cart according to the embodiment.
- [0031] FIG. **18** is a control block diagram of the operation unit according to the embodiment.
- [0032] FIG. **19** is a control flow diagram of a first controller for an operation to rotate a shaft out of its rotational operating range.
- [0033] FIG. **20** is a diagram showing a state in which indicators indicating the movable ranges of the robot arms, the movable ranges of the operation units, and the current positions of the robot arms are displayed on a monitor and a display.
- [0034] FIG. **21** is a diagram showing the indicator indicating the movable range of the robot arm, the movable range of the operation unit, and the current position of the robot arm.
- [0035] FIG. **22** is a diagram showing a state in which a message is displayed, indicating that the operation is outside the rotational operating range of the shaft.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

Configuration of Robotic Surgical System

- [0036] The configuration of a robotic surgical system **500** according to this embodiment is now described. The robotic surgical system **500** includes a surgical robot **100**, a remote control apparatus **200**, a vision unit **300**, and an image processing unit **400**. The remote control apparatus **200** is an example of an operation apparatus.
- [0037] In this specification, as shown in FIG. **4**, the longitudinal direction of a surgical instrument **1** is defined as a Z direction. The distal end side of the surgical instrument **1** is defined as a Z1 side, and the proximal end side of the surgical instrument **1** is defined as a Z2 side. A direction perpendicular to the Z direction is defined as an X direction. A direction perpendicular to the Z direction and the X direction is defined as a Y direction.
- [0038] As shown in FIG. **1**, the surgical robot **100** is arranged in an operating room. The remote control apparatus **200** is spaced apart from the surgical robot **100**. The remote control apparatus **200** receives operations for the surgical instrument **1**. Specifically, an operator such as a doctor

inputs a command to the remote control apparatus **200** to cause the surgical robot **100** to perform a desired operation. The remote control apparatus **200** transmits the input command to the surgical robot **100**. The surgical robot **100** operates based on the received command. The surgical robot **100** is arranged in the operating room that is a sterilized sterile field.

Configuration of Surgical Robot

[0039] As shown in FIG. **1**, the surgical robot **100** includes a medical cart **10**, a cart positioner operation unit **20**, a positioner **30**, an arm base **40**, a plurality of robot arms **50**, and an arm operation unit **60** provided on each of the robot arms **50**.

[0040] As shown in FIG. **3**, the cart positioner operation unit **20** is supported by a cart positioner operation support **21** at the rear of the medical cart **10**, and the medical cart **10** or the positioner **30** is moved by operating the cart positioner operation unit **20**. The cart positioner operation unit **20** includes an input **22** and an operation handle **23**. The input **22** receives operations to move the positioner **30**, the arm base **40**, and the plurality of robot arms **50** or change their postures mainly in order to prepare for surgery before the surgery. The medical cart **10** includes the operation handle **23**, and a stabilizer **24** and an electric cylinder **25** shown in FIG. **15**.

[0041] As shown in FIG. **3**, the input **22** of the medical cart **10** includes a display **22a**, a joystick **22b**, an enable switch **22c**, an error reset button **22d**, and speakers **22e**. The display **22a** is a liquid crystal panel, for example. As shown in FIG. **2**, the display **22a** displays numbers corresponding to the plurality of robot arms **50**. The display **22a** also displays the type of surgical instrument **1** attached to each of the plurality of robot arms **50**. A check mark CM indicating that a pivot position PP described below has been set is displayed on the display **22a**.

[0042] As shown in FIG. **3**, the joystick **22b** is arranged in the vicinity of or adjacent to the display **22a** of the input **22** of the medical cart **10**. The positioner **30** is moved three-dimensionally by selecting an operation mode displayed on the display **22a** and operating the joystick **22b**.

[0043] The enable switch **22c** is arranged in the vicinity of or adjacent to the joystick **22b**. The enable switch **22c** enables or disables movement of the positioner **30**. When the joystick **22b** is operated while the enable switch **22c** is pressed to enable movement of the positioner **30**, the positioner **30** is moved.

[0044] The error reset button **22d** resets errors in the robotic surgical system **500**. The errors may be abnormal deviation errors, for example. The speakers **22e** are arranged in pair. The pair of speakers **22e** are arranged in the vicinity of or adjacent to the location of the positioner **30** in the medical cart **10**.

[0045] The operation handle **23** is arranged in the vicinity of the display **22a**. The operation handle **23** includes a throttle **23a** that is gripped and twisted by an operator such as a nurse or a technician to operate movement of the medical cart **10**. Specifically, the operation handle **23** is arranged below the input **22**. As the throttle **23a** is twisted from the near side to the far side, the medical cart **10** moves forward. As the throttle **23a** is twisted from the far side to the near side, the medical cart **10** moves rearward. The speed of the medical cart **10** is changed according to a twisting amount of the throttle **23a**. The operation handle **23** is rotatable to the left and right shown by an R direction, and the medical cart **10** is turned with rotation of the operation handle **23**.

[0046] An enable switch **23b** for enabling or disabling movement of the medical cart **10** is provided on the operation handle **23** of the cart positioner operation unit **20**. When the throttle **23a** of the operation handle **23** is operated while the enable switch **23b** is pressed to enable movement of the medical cart **10**, the medical cart **10** is moved.

[0047] As shown in FIG. **1**, the positioner **30** includes a 7-axis articulated robot, for example. The positioner **30** is arranged on the medical cart **10**. The positioner **30** adjusts the position of the arm base **40**. The positioner **30** moves the position of the arm base **40** three-dimensionally.

[0048] The positioner **30** includes a base **31** and a plurality of links **32** coupled to the base **31**. The plurality of links **32** are coupled to each other by joints **33**.

[0049] The arm base **40** is attached to the distal end of the positioner **30**. The proximal end of each

of the plurality of robot arms **50** is attached to the arm base **40**. Each of the plurality of robot arms **50** is able to take a folded and stored posture. The arm base **40** and the plurality of robot arms **50** are covered with sterile drapes and used. Moreover, each of the robot arms **50** supports the surgical instrument **1**.

[0050] A status indicator **41** and an arm status indicator **42** that are shown in FIG. **15** are provided on the arm base **40**. The status indicator **41** indicates the status of the robotic surgical system **500**. The arm status indicator **42** indicates the statuses of the robot arms **50**.

[0051] The plurality of robot arms **50** are arranged. Specifically, four robot arms **50a**, **50b**, **50c**, and **50d** are arranged. The robot arms **50a**, **50b**, **50c**, and **50d** have the same or similar configurations as each other.

[0052] As shown in FIG. **4**, each robot arm **50** includes an arm portion **51**, a first link **52**, a second link **53**, and a translation mechanism **54**. The robot arm **50** includes joints JT1, JT2, JT3, JT4, JT5, JT6, JT7, and JT8. The joints JT1, JT2, JT3, JT4, JT5, JT6, and JT7 have A1, A2, A3, A4, A5, A6, and A7 axes as rotation axes, respectively. The joint JT8 has an A8 axis as a linear motion axis. The arm portion **51** includes a base **51a** and links **51b**.

[0053] The arm portion **51** includes a 7-axis articulated robot arm. The first link **52** is arranged at the distal end of the arm portion **51**. The arm operation unit **60** described below is attached to the second link **53**. The translation mechanism **54** is arranged between the first link **52** and the second link **53**. A holder **55** that holds the surgical instrument **1** is arranged on the second link **53**. The translation mechanism **54** translates the holder **55** to which the surgical instrument **1** is attached between a first position and a second position. The first position refers to an end position on the Z2 side in a range of movement of the holder **55** along the A8 axis by the translation mechanism **54**. The second position refers to an end position on the Z1 side in the range of movement of the holder **55** along the A8 axis by the translation mechanism **54**.

[0054] The surgical instrument **1** is attached to the distal end of each of the plurality of robot arms **50**. The surgical instrument **1** includes a replaceable instrument **2**, an endoscope **3** shown in FIG. **9** to capture an image of a surgical site, and a pivot position setting instrument **4** shown in FIG. **10** to set the pivot position PP, for example. The instrument **2** includes a driven unit **2a**, an end effector **2b**, and a wrist joint **2c** and a shaft **2d** shown in FIG. **5**. The end effector **2b** is connected to the distal end of the shaft **2d** via the wrist joint **2c**. The instrument **2** and the end effector **2b** are examples of a surgical instrument and a distal end device, respectively.

[0055] As shown in FIG. **1**, the endoscope **3** is attached to the distal end of one of the plurality of robot arms **50**, such as the robot arm **50c**, and the instrument **2** is attached to the distal end of each of the remaining robot arms **50a**, **50b**, and **50d**, for example. The endoscope **3** is preferably attached to one of two robot arms **50b** and **50c** arranged in the center among the four robot arms **50** arranged adjacent to each other.

Configuration of Instrument

[0056] As shown in FIG. **5**, the end effector **2b** including jaw members **2g** and **2h** is attached to the distal end of the instrument **2**, for example. As the end effector **2b**, a pair of scissors, a grasper, a needle holder, a microdissector, a stable applier, a tacker, a suction cleaning tool, a snare wire, a clip applier, etc. can be used.

[0057] The instrument **2** includes a first support member **2e** and a second support member **2f**. The first support member **2e** is attached to the shaft **2d**. The second support member **2f** is supported by the first support member **2e** so as to be rotatable around an A10 axis, and supports the end effector **2b** such that the end effector **2b** is rotatable around an A11 axis that intersects with the A10 axis as viewed in a direction in which the shaft **2d** extends. The shaft **2d** rotates around an A9 axis. The wrist joint **2c** is provided between the second support member **2f** and the first support member **2e** with the A10 axis as a rotation axis. The A10 axis and the A11 axis are examples of a first rotation axis and a second rotation axis, respectively.

Configuration of Arm Operation Unit

[0058] As shown in FIG. 6, the arm operation unit **60** is attached to the robot arm **50** to operate the robot arm **50**. Specifically, the arm operation unit **60** is attached to the second link **53**.

[0059] The arm operation unit **60** includes an enable switch **61**, a joystick **62**, and linear switches **63**, a mode switching button **64**, a mode indicator **65**, a pivot button **66**, and an adjustment button **67**.

[0060] The enable switch **61** is pressed to enable or disable movement of the robot arm **50** in response to the joystick **62** and the linear switches **63**. When the enable switch **61** is pressed by an operator such as a nurse or an assistant grasping the arm operation unit **60**, movement of the surgical instrument **1** by the robot arm **50** is enabled.

[0061] The joystick **62** is an operation tool to control movement of the surgical instrument **1** by the robot arm **50**. The joystick **62** controls a moving direction and a moving speed of the robot arm **50**. The robot arm **50** is moved in accordance with a tilting direction and a tilting angle of the joystick **62**.

[0062] The linear switches **63** are switches to move the surgical instrument **1** in the Z direction, which is the longitudinal direction of the surgical instrument **1**. The linear switches **63** include a linear switch **63a** to move the surgical instrument **1** in a direction in which the surgical instrument **1** is inserted into a patient P, and a linear switch **63b** to move the surgical instrument **1** in a direction in which the surgical instrument **1** is moved away from the patient P. Both the linear switch **63a** and the linear switch **63b** are push-button switches.

[0063] The mode switching button **64** is a push-button switch to switch between a mode for translationally moving the surgical instrument **1** and a mode for rotationally moving the surgical instrument **1**. As shown in FIG. 7, in the mode for translationally moving the robot arm **50**, the robot arm **50** is moved such that the distal end **1a** of the surgical instrument **1** is moved in an X-Y plane. As shown in FIG. 8, in the mode for rotationally moving the robot arm **50**, the robot arm **50** is moved such that the surgical instrument **1** is rotationally moved around a center on the A11 axis of the end effector **2b** or the distal end of the end effector **2b** of the instrument **2** as the surgical instrument **1** as a fulcrum when any pivot position PP is not stored in a storage **351**, and the surgical instrument **1** is rotationally moved around the pivot position PP as a fulcrum when the pivot position PP is stored in the storage **351**. In this case, the surgical instrument **1** is rotationally moved with the shaft **1c** of the surgical instrument **1** inserted into a trocar T. The mode switching button **64** is arranged on a Z-direction side surface of the arm operation unit **60**.

[0064] The mode indicator **65** indicates a switched mode. The mode indicator **65** is on to indicate a rotational movement mode and is off to indicate a translational movement mode. Furthermore, the mode indicator **65** also serves as a pivot position indicator that indicates that the pivot position PP has been set. The mode indicator **65** is arranged on the Z-direction side surface of the arm operation unit **60**.

[0065] The pivot button **66** is a push-button switch to set the pivot position PP that serves as a fulcrum for movement of the surgical instrument **1** attached to the robot arm **50**.

[0066] The adjustment button **67** is a button to optimize the position of the robot arm **50**. After the pivot position PP for the robot arm **50** to which the endoscope **3** has been attached is set, the positions of the other robot arms **50** and the arm base **40** are optimized when the adjustment button **67** is pressed. The adjustment button **67** is a button different from the enable switch **61**.

Remote Control Apparatus

[0067] As shown in FIG. 1, the remote control apparatus **200** is arranged inside or outside the operating room, for example. The remote control apparatus **200** includes the operation units **110**, foot pedals **120**, a touch panel **130**, a monitor **140**, a support arm **150**, a support bar **160**, an error reset button **161**. The operation units **110** include operation handles for the operator such as a doctor to input a command. The monitor **140** is an example of a first display or a second display.

Operation Unit

[0068] As shown in FIG. 11, the operation units **110** each include a handle to operate the surgical

instrument **1**. The operation unit **110** receives an operation for the surgical instrument **1**. The operation units **110** include an operation unit **110L** that is located on the left side as viewed from the operator such as a doctor and is to be operated by the left hand of the operator, and an operation unit **110R** that is located on the right side and is to be operated by the right hand of the operator. The operation units **110** include arms **111** and wrists **112**. The operation unit **110R** includes an arm **111R** and a wrist **112R**. The operation unit **110L** includes an arm **111L** and a wrist **112L**.

[0069] The operation units **110** each have joints JT21, JT22, and JT23 shown in FIG. **11**, and joints JT24, JT25, JT26, and JT27 shown in FIGS. **12** and **13**. The rotation axes of the joints JT21, JT22, JT23, JT24, JT25, JT26, and JT27 are A21, A22, A23, A24, A25, A26, and A27 axes, respectively.

[0070] As shown in FIG. **11**, the arm **111R** includes a link **111a**, a link **111b**, and a link **111c**. The upper end side of the link **111a** is attached to the remote control apparatus **200** such that the link **111a** is rotatable around the A21 axis along a vertical direction. The upper end side of the link **111b** is attached to the lower end side of the link **111a** such that the link **111b** is rotatable around the A22 axis along a horizontal direction. A first end side of the link **111c** is attached to the lower end side of the link **111b** such that the link **111c** is rotatable around the A23 axis along the horizontal direction. The wrist **112** is attached to a second end side of the link **111c** such that the wrist **112** is rotatable around the A24 axis. The link **111a** is connected to the remote control apparatus **200** by the joint JT21. The link **111a** and the link **111b** are connected to each other by the joint JT22. The link **111b** and the link **111c** are connected to each other by the joint JT23. The arm **111** supports the wrist **112**. The arm **111L** has the same or similar configuration as the arm **111R**.

[0071] The wrists **112** include the wrist **112R** shown in FIG. **12** and to be operated by the right hand of the operator, and the wrist **112L** shown in FIG. **13** and to be operated by the left hand of the operator. FIG. **12** shows the reference posture of the operation unit **110R**, and FIG. **13** shows the reference posture of the operation unit **110L**. The configuration of the wrist **112R** is the same as or similar to that of the wrist **112L**.

[0072] The wrists **112** each include a link **112a**, a link **112b**, a link **112c**, and a grip support member **112d** that is to be operated by the operator such as a doctor. The link **112a** includes a proximal end connected to the distal end of the arm **111** and rotates around the A24 axis. The link **112b** includes a proximal end connected to the distal end of the link **112a** and rotates around the A25 axis with respect to the link **112a**. The link **112c** includes a proximal end connected to the distal end of the link **112b** and a distal end to which the grip support member **112d** is connected, and rotates around the A26 axis with respect to the link **112b**. The grip support member **112d** rotates around the A27 axis with respect to the link **112c**. The link **112a**, the link **112b**, and the link **112c** each have an L shape. The link **112a**, the link **112b**, and the link **112c** are examples of a first link, a second link, and a third link, respectively.

[0073] The wrist **112** includes a pair of grip members **112e** that are opened and closed by the operator. The grip members **112e** each include an elongated plate-shaped lever member, and the proximal ends of the pair of grip members **112e** are rotatably connected to the proximal end of the grip support member **112d**. Cylindrical finger insertion portions **112f** are provided on the grip members **112e**. The operator inserts their fingers into a pair of finger insertion portions **112f** to operate the wrist **112**. The proximal ends of the pair of grip members **112e** are connected to the grip support member **112d**, and an angle between the pair of grip members **112e** is increased or decreased such that an opening angle between the jaw member **2g** and the jaw member **2h** is changed. A magnet is provided on one of the grip members **112e**, and a Hall sensor is provided on the grip support member **112d**. When the operator opens and closes the grip members **112e**, the magnet and the Hall sensor function as an angle detection sensor, and the Hall sensor outputs the opening angle. As the angle detection sensor, the Hall sensor may be provided on the grip member **112e**, and the magnet may be provided on the grip support member **112d**. Alternatively, the magnet or the Hall sensor may be provided as the angle detection sensor on both the grip members **112e**.

[0074] As shown in FIG. **1**, the monitor **140** is a scope-type display that displays images captured

by the endoscope **3**. A notifier **141** is provided on the monitor **140**. The notifier **141** issues an error sound. The support arm **150** supports the monitor **140** so as to align the height of the monitor **140** with the height of the face of the operator such as a doctor. The touch panel **130** is arranged on the support bar **160**. The head of the operator is detected by a sensor provided in the vicinity of the monitor **140** such that the surgical robot **100** can be operated by the remote control apparatus **200**. The operator operates the operation units **110** and the foot pedals **120** while visually recognizing an affected area on the monitor **140**. Thus, a command is input to the remote control apparatus **200**. The command input to the remote control apparatus **200** is transmitted to the surgical robot **100**.

Foot Pedals

[0075] As shown in FIG. **14**, a plurality of foot pedals **120** are provided to perform functions related to the surgical instrument **1**. The plurality of foot pedals **120** are arranged on a base **121**. The foot pedals **120** include a switching pedal **122**, a clutch pedal **123**, a camera pedal **124**, incision pedals **125**, coagulation pedals **126**, and foot detectors **127**. The switching pedal **122**, the clutch pedal **123**, the camera pedal **124**, the incision pedals **125**, and the coagulation pedals **126** are operated by the foot of the operator. The incision pedals **125** include an incision pedal **125R** for a right robot arm **50**, and an incision pedal **125L** for a left robot arm **50**. The coagulation pedals **126** include a coagulation pedal **126R** for the right robot arm **50** and a coagulation pedal **126L** for the left robot arm **50**.

[0076] The switching pedal **122** switches robot arms **50** to be operated by the operation units **110**. The clutch pedal **123** performs a clutch operation to temporarily disconnect an operation connection between the robot arms **50** and the operation units **110**. While the clutch pedal **123** is being pressed by the operator, operations by the operation units **110** are not transmitted to the robot arms **50**. While the camera pedal **124** is being pressed by the operator, the operation unit **110** can operate a robot arm **50** to which the endoscope **3** is attached. While the incision pedals **125** or the coagulation pedals **126** are being pressed by the operator, an electrosurgical device is activated.

Vision Unit and Image Processing Unit

[0077] As shown in FIG. **1**, the vision unit **300** and the image processing unit **400** are placed on a cart **210**. The image processing unit **400** processes images captured by the endoscope **3**. A display **220** is arranged on the cart **210**. The display **220** displays images captured by the endoscope **3**. The display **220** is an example of a first display or a second display.

Configuration of Control System

[0078] As shown in FIG. **15**, the robotic surgical system **500** includes a first control device **310**, an arm controller **320**, a positioner controller **330**, operation controllers **340**, and a second control device **350**. The robotic surgical system **500** also includes a storage **311** connected to the first control device **310** and the storage **351** connected to the second control device **350**. The first control device **310** is an example of a controller.

[0079] The first control device **310** is accommodated in the medical cart **10** to communicate with the arm controller **320** and the positioner controller **330**, and controls the entire robotic surgical system **500**. Specifically, the first control device **310** communicates with and controls the arm controller **320**, the positioner controller **330**, and the operation controllers **340**. The first control device **310** is connected to the arm controller **320**, the positioner controller **330**, and the operation controllers **340** through a LAN, for example. The first control device **310** is arranged inside the medical cart **10**.

[0080] The arm controller **320** is arranged for each of the plurality of robot arms **50**. That is, the same number of arm controllers **320** as the plurality of robot arms **50** are placed inside the medical cart **10**.

[0081] As shown in FIG. **15**, the input **22** is connected to the first control device **310** through a LAN, for example. The status indicator **41**, the arm status indicator **42**, the operation handle **23**, the throttle **23a**, the joystick **22b**, the stabilizer **24**, and the electric cylinder **25** are connected to the positioner controller **330** via a wire line **360** by means of a communication network that allows

information to be shared with each other by using serial communication. Although FIG. 15 shows that the status indicator **41**, the arm status indicator **42**, etc. are all connected to one wire line **360**, in reality, the wire line **360** is arranged for each of the status indicator **41**, the arm status indicator **42**, the operation handle **23**, the throttle **23a**, the joystick **22b**, the stabilizer **24**, and the electric cylinder **25**.

[0082] As shown in FIG. 16, the arm portion **51** includes a plurality of servomotors **SM1**, encoders **EN1**, and speed reducers so as to correspond to the joints **JT1**, **JT2**, **JT3**, **JT4**, **JT5**, **JT6**, and **JT7**. The encoders **EN1** detect rotation angles of the servomotors **SM1**. The speed reducers slow down rotation of the servomotors **SM1** to increase the torques. Inside the medical cart **10**, servo controllers **SC1** that control the servomotors **SM1** are arranged adjacent to the arm controller **320**. In addition, the encoders **EN1** that detect the rotation angles of the servomotors **SM1** are electrically connected to the servo controllers **SC1**.

[0083] Servomotors **SM2** that rotate driven members provided in the driven unit **2a** of the surgical instrument **1**, encoders **EN2**, and speed reducers are arranged in the second link **53**. The encoders **EN2** detect rotation angles of the servomotors **SM2**. The speed reducers slow down rotation of the servomotors **SM2** to increase the torques. In the medical cart **10**, servo controllers **SC2** are provided to control the servomotors **SM2** to drive the surgical instrument **1**. The encoders **EN2** that detect the rotation angles of the servomotors **SM2** are electrically connected to the servo controllers **SC2**. A plurality of servomotors **SM2**, a plurality of encoders **EN2**, and a plurality of servo controllers **SC2** are arranged.

[0084] The translation mechanism **54** includes a servomotor **SM3** to translationally move the surgical instrument **1**, an encoder **EN3**, and a speed reducer. The encoder **EN3** detects a rotation angle of the servomotor **SM3**. The speed reducer slows down rotation of the servomotor **SM3** to increase the torque. In the medical cart **10**, a servo controller **SC3** is provided to control the servomotor **SM3** to translationally move the surgical instrument **1**. The encoder **EN3** that detects the rotation angle of the servomotor **SM3** is electrically connected to the servo controller **SC3**.

[0085] The first control device **310** generates command values to command the positions of the servomotors **SM1**, **SM2**, and **SM3** based on operations received by the remote control apparatus **200**, and drives the servomotors **SM1**, **SM2**, and **SM3** based on the command values. The first control device **310** detects an abnormal deviation error when differences between the command values and the positions of the servomotors **SM1**, **SM2**, and **SM3** detected by sensors exceed an allowable range.

[0086] As shown in FIG. 17, a plurality of servomotors **SM4**, a plurality of encoders **EN4**, and a plurality of speed reducers are provided in the positioner **30** so as to correspond to the plurality of joints **33** of the positioner **30**. The encoders **EN4** detect rotation angles of the servomotors **SM4**. The speed reducers slow down rotation of the servomotors **SM4** to increase the torques.

[0087] The medical cart **10** includes wheels including front wheels as drive wheels and rear wheels that are steered by the operation handle **23**. The rear wheels are arranged closer to the operation handle **23** than the front wheels. The medical cart **10** includes servomotors **SM5** to drive a plurality of front wheels of the medical cart **10**, encoders **EN5**, speed reducers, and brakes **BRK**. The speed reducers slow down rotation of the servomotors **SM5** to increase the torques. A potentiometer **P1** shown in FIG. 3 is provided on the operation handle **23** of the medical cart **10**, and the servomotors **SM5** of the front wheels are driven based on a rotation angle detected by the potentiometer **P1** according to the twist of the throttle **23a**. Rear wheels of the medical cart **10** are of the dual wheel type, and the rear wheels are steered based on rightward-leftward rotation of the operation handle **23**. Furthermore, a potentiometer **P2** shown in FIG. 3 is provided on a rotation axis of the operation handle **23** of the medical cart **10**, and servomotors **SM6**, encoders **EN6**, and speed reducers are provided on the rear wheels of the medical cart **10**. The speed reducers slow down rotation of the servomotors **SM6** to increase the torques. The servomotors **SM6** are driven based on a rotation angle detected by the potentiometer **P2** according to rightward-leftward rotation of the operation

handle **23**. That is, steering of the rear wheels by the rightward-leftward rotation of the operation handle **23** is power-assisted by the servomotors **SM6**.

[0088] The front wheels of the medical cart **10** are driven such that the medical cart **10** moves in a forward-rearward direction. Furthermore, the operation handle **23** of the medical cart **10** is rotated such that the rear wheels are steered, and the medical cart **10** turns in a right-left direction.

[0089] As shown in FIG. **17**, in the medical cart **10**, servo controllers **SC4** are provided to control the servomotors **SM4** to move the positioner **30**. The encoders **EN4** that detect the rotation angles of the servomotors **SM4** are electrically connected to the servo controllers **SC4**. In the medical cart **10**, servo controllers **SC5** are provided to control the servomotors **SM5** to drive the front wheels of the medical cart **10**. The encoders **EN5** that detect the rotation angles of the servomotors **SM5** are electrically connected to the servo controllers **SC5**. In the medical cart **10**, servo controllers **SC6** are provided to control the servomotors **SM6** to power-assist steering of the rear wheels of the medical cart **10**. The encoders **EN6** that detect the rotation angles of the servomotors **SM6** are electrically connected to the servo controllers **SC6**.

[0090] As shown in FIGS. **16** and **17**, the joints **JT1**, **JT2**, **JT3**, **JT4**, **JT5**, **JT6**, and **JT7** of the arm portion **51**, and the joints **33** of the positioner **30** include brakes **BRK**. Furthermore, the front wheels of the medical cart **10**, the arm base **40**, and the translation mechanism **54** include brakes **BRK**. The arm controller **320** unidirectionally transmits control signals to the brakes **BRK** of the joints **JT1**, **JT2**, **JT3**, **JT4**, **JT5**, **JT6**, and **JT7** of the arm portion **51**, and the translation mechanism **54**. The control signals are signals for turning on/off the brakes **BRK**. The signals for turning on the brakes **BRK** include signals for maintaining the brakes **BRK** in an enabled state. The same applies to control signals from the positioner controller **330** to the brakes **BRK** of the joints **33** of the positioner **30** and the arm base **40**. On startup, all the brakes **BRK** of the arm base **40**, the arm portion **51**, and the translation mechanism **54** are turned off but the servomotors **SM** are driven against gravity to maintain the postures of the robot arm **50** and the arm base **40**. When an error occurs in the robotic surgical system **500**, the brakes **BRK** of the arm base **40**, the arm portion **51** and the translation mechanism **54** are turned on. When the error in the robotic surgical system **500** is reset, the brakes **BRK** of the arm base **40**, the arm portion **51**, and the translation mechanism **54** are turned off. When a shutdown operation is performed in the robotic surgical system **500**, the brakes **BRK** of the arm base **40**, the arm portion **51**, and the translation mechanism **54** are turned on. The brakes **BRK** of the front wheels of the medical cart **10** are constantly turned on, and the brakes **BRK** are deactivated only while the enable switch **23b** of the medical cart **10** is being pressed. The brakes **BRK** of the joints **33** of the positioner **30** are constantly turned on, and the brakes **BRK** are deactivated only while the enable switch **22c** of the medical cart **10** is being pressed.

[0091] As shown in FIG. **18**, servomotors **SM7a**, **SM7b**, **SM7c**, **SM7d**, **SM7e**, **SM7f**, and **SM7g** are arranged on the joints **JT21**, **JT22**, **JT23**, **JT24**, **JT25**, **JT26**, and **JT27** of the operation unit **110**, respectively. The servomotor **SM7a** rotates the link **111a** around the **A21** axis. The servomotor **SM7b** rotates the link **111b** around the **A22** axis. The servomotor **SM7c** rotates the link **111c** around the **A23** axis. The servomotor **SM7d** rotates the link **112a** around the **A24** axis. The servomotor **SM7e** rotates the link **112b** around the **A25** axis. The servomotor **SM7f** rotates the link **112c** around the **A26** axis. The servomotor **SM7g** rotates the grip support member **112d** around the **A27** axis. The **A24** axis, the **A25** axis, and the **A26** axis are examples of a fourth rotation axis, a fifth rotation axis, and a sixth rotation axis, respectively. The **A27** axis is an example of a third rotation axis. Servo controllers **SC7a**, **SC7b**, **SC7c**, **SC7d**, **SC7e**, **SC7f**, and **SC7g** are provided to control the servomotors. Encoders **EN7a**, **EN7b**, **EN7c**, **EN7d**, **EN7e**, **EN7f**, and **EN7g** are electrically connected to the servo controllers to detect rotation angles of the servomotors. The servomotors, the servo controllers, and the encoders are provided in each of the operation unit **110L** and the operation unit **110R**. The servomotors **SM7d**, **SM7e**, and **SM7f** are examples of a first link motor, a second link motor, and a third link motor, respectively. The servomotor **SM7g** is an example of a

grip support member motor.

[0092] The first control device **310** controls the servomotors via the operation controllers **340** to generate torques that counteract gravitational torques generated on rotation axes of the servomotors according to the postures of the operation units **110**. Thus, the operator can operate the operation units **110** with a relatively small force.

[0093] When the operator performs an operation to rotate the grip support member **112d** around the A27 axis of the operation unit **110** shown in FIGS. **12** and **13**, the shaft **2d** of the instrument **2** rotates around the A9 axis shown in FIG. **5**. When the operator performs an operation to rotate the joints JT24, JT25, and JT26 of the operation unit **110** shown in FIGS. **12** and **13**, the end effector **2b** bends around the A10 axis or the A11 axis shown in FIG. **5**. The shaft **2d** has a rotational operating range around the A9 axis. The rotational operating range is defined, for example, as a rotatable angle from $+A^{\circ}$ to $-A^{\circ}$. The rotational operating range of the shaft **2d** includes a mechanical rotational operating range determined based on a machine constituting the shaft **2d** and a rotational operating range defined on software preset in the robotic surgical system **500**. In this embodiment, the rotational operating range of the shaft **2d** defined on the software is smaller than the mechanical rotational operating range.

[0094] The operation controllers **340** are arranged in a main body of the remote control apparatus **200**. The operation controllers **340** control the operation units **110**. The operation controllers **340** are provided so as to correspond to the left-handed operation unit **110L** and the right-handed operation unit **110R**, respectively, as shown in FIG. **15**.

[0095] As shown in FIG. **15**, the vision unit **300** and the image processing unit **400** are connected to the first control device **310** through a LAN, for example. The display **220** is connected to the vision unit **300**.

Control for Operation Out of Rotational Operating Range of Shaft

[0096] The control of the first control device **310** in a case in which an operation for the instrument **2** received by the operation unit **110** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d** is now described with reference to FIG. **19**.

[0097] In step S1, the first control device **310** starts to receive an operation for the instrument **2** with the operation unit **110**. In this embodiment, as shown in FIG. **20**, the monitor **140** of the remote control apparatus **200** and the display **220** arranged on the cart **210** display an indicator M indicating the movable range M1 of the robot arm **50**, the movable range M2 of the operation unit **110** with respect to the movable range M1 of the robot arm **50**, and the current position M3 of the robot arm **50**. The indicator M is displayed for each of the left-handed operation unit **110L** and the right-handed operation unit **110R**. For example, in an example shown in FIG. **21**, the horizontal length of the movable range M2 of the operation unit **110** is smaller than the horizontal length of the movable range M1 of the robot arm **50**, and the example indicates that the movable range M2 of the operation unit **110** is smaller than the movable range M1 of the robot arm **50**. Furthermore, the current position M3 of the robot arm **50** has an inverted triangle shape, for example, and the position indicated by the inverted triangle shape represents the current position M3 of the robot arm **50**.

[0098] In step S2, the operator performs an operation to rotate the grip support member **112d** around the A27 axis of the operation unit **110** shown in FIG. **12** or **13**, and the first control device **310** receives an operation to rotate the shaft **2d** of the instrument **2** around the A9 axis shown in FIG. **5**.

[0099] In step S3, in this embodiment, the first control device **310** determines whether or not the operation for the instrument **2** received by the operation unit **110** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**. Specifically, the first control device **310** determines whether or not the operation for the instrument **2** received by the grip support member **112d** of the operation unit **110** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**. In this embodiment, a rotational operating range that

serves as a criterion for the determination of the first control device **310** is a rotational operating range defined on the software that is smaller than the mechanical rotational operating range of the shaft **2d**. Furthermore, the operation to rotate the shaft **2d** received by the grip support member **112d** of the operation unit **110** refers to an operation to rotate the grip support member **112d** around the A27 axis. For example, when the rotatable angle is defined as $+A^\circ$ to $-A^\circ$, and the operator attempts to rotate the grip support member **112d** around the A27 axis to rotate the shaft **2d** to an angle that exceeds an angular range from $+A^\circ$ to $-A^\circ$, the first control device **310** determines yes in step S3.

[0100] In a case of yes in step S3, in this embodiment, in step S4, the first control device **310** restricts an operation to rotate the shaft **2d** in a direction out of the rotational operating range with the operation unit **110** and an operation to bend the end effector **2b** with respect to the shaft **2d** with the operation unit **110**. Specifically, when the operation for the instrument **2** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**, the first control device **310** controls the servomotor SM7g to prevent rotation of the grip support member **112d**. For example, when the rotatable angle of the shaft **2d** is defined as $+A^\circ$ to $-A^\circ$, and the operator attempts to rotate the grip support member **112d** around the A27 axis to rotate the shaft **2d** to an angle greater than $+A^\circ$, the first control device **310** transmits a command to the servo controller SC7g to maintain the rotation angle of the shaft **2d** at $+A^\circ$. Thus, even when the operator attempts to rotate the grip support member **112d** to rotate the shaft **2d** in a direction greater than $+A'$, a relatively large load is applied to the grip support member **112d** such that it is difficult for the grip support member **112d** to rotate. In other words, when the operator attempts to rotate the grip support member **112d** to rotate the shaft **2d** in a direction greater than $+A^\circ$, the servomotor SM7g generates a force to return the grip support member **112d** to an angle of $+A^\circ$ such that it is difficult for the grip support member **112d** to rotate. On the other hand, when the grip support member **112d** is rotated to rotate the shaft **2d** in a direction smaller than $+A^\circ$, a load applied to the grip support member **112d** is small such that the operator can easily rotate the grip support member **112d**.

[0101] In this embodiment, in a case of yes in step S3, in step S4, an operation to bend the end effector **2b** with respect to the shaft **2d** with the link **112a**, the link **112b**, and the link **112c** is restricted. The operation to bend the end effector **2b** with respect to the shaft **2d** includes an operation to rotate the second support member **2f** around the A10 axis and an operation to rotate the end effector **2b** around the A11 axis. Specifically, when the operation for the instrument **2** received by the operation unit **110** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**, the first control device **310** restricts rotation of the link **112a** around the A24 axis, rotation of the link **112b** around the A25 axis, and rotation of the link **112c** around the A26 axis. More specifically, when the operation for the instrument **2** received by the operation unit **110** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**, the first control device **310** controls the servomotors SM7d, SM7e, and SM7f to prevent rotation of the links **112a**, **112b**, and **112c**. For example, in a case of yes in step S3, the first control device **310** transmits commands to the servo controllers SC7d, SC7e, and SC7f to maintain the rotation angles of the links **112a**, **112b**, and **112c** at the angles at the time at which the determination of “yes” is made in step 103. Thus, even when the operator attempts to rotate the links **112a**, **112b**, and **112c**, a relatively large load is applied to the links **112a**, **112b**, and **112c** such that it is difficult for the links **112a**, **112b**, and **112c** to rotate. In other words, in a case of yes in step S3, the servomotors SM7d, SM7e, and SM7f generate forces to return the links **112a**, **112b**, and **112c** to the angles at the time at which the determination of “yes” is made in step 103 such that it is difficult for the links **112a**, **112b**, and **112c** to rotate.

[0102] Even in step S4 in which the operation for the instrument **2** received by the operation unit **110** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**, an operation on the arm **111** of the operation unit **110** is continuously received. Furthermore, an operation to open and close the jaw members **2g** and **2h** of the instrument **2** by opening and closing

the grip members **112e** of the operation unit **110** is received.

[0103] As shown in FIG. **21**, in a case of yes in step **S3**, the current position **M3** of the robot arm **50** shown on the monitor **140** of the remote control apparatus **200** and the display **220** arranged on the cart **210** has moved to an end of the movable range **M2** of the operation unit **110**. By visually checking the monitor **140** or the display **220**, the operator can visually recognize that the limit of the movable range **M2** of the operation unit **110** has been reached.

[0104] In this embodiment, in step **S5**, as shown in FIG. **22**, when the operation for the instrument **2** received by the operation unit **110** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**, the monitor **140** of the remote control apparatus **200** and the display **220** arranged on the cart **210** display a message indicating that the operation on the operation unit **110** is an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**. For example, the first control device **310** executes a process to display the message that the operation is an operation to rotate the shaft **2d** out of the rotational operating range on the monitor **140** of the remote control apparatus **200** and the display **220** arranged on the cart **210**.

[0105] In a case of no in step **S3**, the process returns to step **S1**. The operations from step **S1** to step **S5** are repeated while an operation with the operation unit **110** is being received.

Advantages of this Embodiment

[0106] The first control device **310** is configured or programmed to restrict an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d** with the operation unit **110** and an operation to bend the end effector **2b** with respect to the shaft **2d** with the operation unit **110** when the operation for the instrument **2** received by the operation unit **110** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**. Accordingly, even when an operation is performed to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**, an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d** with the operation unit **110** and an operation to bend the end effector **2b** with respect to the shaft **2d** with the operation unit **110** are restricted, and thus the posture of the operation unit **110** is maintained.

Therefore, the possibility that the postures of the end effector **2b** and the shaft **2d** do not match the posture of the operation unit **110** is reduced or prevented, and thus a state in which the postures of the end effector **2b** and the shaft **2d** match the posture of the operation unit **110** is maintained. In addition, the first control device **310** only restricts an operation to rotate the shaft **2d** out of the rotational operating range with the operation unit **110** and an operation to bend the end effector **2b** with respect to the shaft **2d** with the operation unit **110**, but does not restrict reception of all operations with the operation unit **110**. Therefore, even when an operation is performed that attempts to exceed the movable range of the instrument **2**, a sudden change in the posture of the instrument **2** can be reduced or prevented while a state in which an operation with the operation unit **110** can be received is maintained.

[0107] The rotational operating range of the shaft **2d** is a rotational operating range defined on software that is smaller than the mechanical rotational operating range of the shaft **2d**. Accordingly, the shaft **2d** is prevented from operating beyond the mechanical rotational operating range, and thus mechanical failure of the instrument **2** can be prevented.

[0108] The instrument **2** includes the first support member **2e** attached to the shaft **2d**, and the second support member **2f** supported by the first support member **2e** so as to be rotatable around the A10 axis to support the end effector **2b** such that the end effector **2b** is rotatable around the A11 axis intersecting with the A10 axis as viewed in the direction in which the shaft **2d** extends. The operation to bend the end effector **2b** with respect to the shaft **2d** includes an operation to rotate the second support member **2f** around the A10 axis and an operation to rotate the end effector **2b** around the A11 axis. Accordingly, when the operation for the instrument **2** received by the operation unit **110** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**, rotation of the second support member **2f** around the A10 axis and rotation of the end effector **2b** around the A11 axis can be restricted.

[0109] The operation unit **110** includes the grip support member **112d** to rotatably support one ends of the grip members **112e** to be operated by the fingers of the operator and operable to rotate around the A27 axis parallel to the longitudinal direction of the grip support member **112d** to receive an operation to rotate the shaft **2d**. An operation to rotate the shaft **2d** of the instrument **2** is received by rotating the grip support member **112d** around the A27 axis. Accordingly, an operation to rotate the grip support member **112d** out of the rotational operating range of the shaft **2d** is restricted, and thus the possibility that the rotation angle of the grip support member **112d** does not match the rotation angle of the shaft **2d** can be reduced or prevented.

[0110] The operation unit **110** includes the servomotor SM7g to rotate the grip support member **112d** around the A27 axis. The first control device **310** is configured or programmed to control the servomotor SM7g to prevent rotation of the grip support member **112d** when the operation for the instrument **2** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**. Accordingly, rotation of the grip support member **112d** can be prevented by the prearranged servomotor SM7g without the need to separately arrange a dedicated brake or the like for preventing rotation of the grip support member **112d**. Consequently, the complexity of the configuration of the robotic surgical system **500** can be reduced or prevented.

[0111] The operation unit **110** includes the arm **111** and the wrist **112**. The wrist **112** includes the link **112a** including the proximal end connected to the distal end of the arm **111** and rotatable around the A24 axis, the link **112b** including the proximal end connected to the distal end of the link **112a** and rotatable around the A25 axis, and the link **112c** including the proximal end connected to the distal end of the link **112b** and the distal end to which the grip support member **112d** is connected and rotatable around the A26 axis. The first control device **310** is configured or programmed to restrict rotation of the link **112a** around the A24 axis, rotation of the link **112b** around the A25 axis, and rotation of the link **112c** around the A26 axis when the operation for the instrument **2** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**. Accordingly, rotation of the links **112a**, **112b**, and **112c** is restricted, and thus the possibility that the postures of the links **112a**, **112b**, and **112c** do not match the posture of the instrument **2** can be reduced or prevented.

[0112] The wrist **112** includes the servomotor SM7d to rotate the link **112a** around the A24 axis, the servomotor SM7e to rotate the link **112b** around the A25 axis, and the servomotor SM7f to rotate the link **112c** around the A26 axis. The first control device **310** is configured or programmed to control the servomotors SM7d, SM7e, and SM7f to prevent rotation of the links **112a**, **112b**, and **112c** when the operation for the instrument **2** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**. Accordingly, rotation of the grip support member **112d** can be prevented by the prearranged servomotors SM7d, SM7e, and SM7f without the need to separately arrange a dedicated brake or the like for preventing rotation of the links **112a**, **112b**, and **112c**. Consequently, the complexity of the configuration of the robotic surgical system **500** can be reduced or prevented.

[0113] The monitor **140** of the remote control apparatus **200** and the display **220** arranged on the cart **210** are operable to display the message indicating that the operation on the operation unit **110** is an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d** when the operation for the instrument **2** received by the operation unit **110** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**. Accordingly, by visually checking the message, the operator can easily recognize that the operation on the operation unit **110** is an operation out of the rotational operating range of the shaft **2d**.

[0114] The monitor **140** of the remote control apparatus **200** and the display **220** arranged on the cart **210** are operable to display the indicator M indicating the movable range M1 of the robot arm **50**, the movable range M2 of the operation unit **110** with respect to the movable range M1 of the robot arm **50**, and the current position M3 of the robot arm **50**. Accordingly, by visually checking the current position M3 of the robot arm **50**, the operator can easily recognize whether or not the

operation on the operation unit **110** has reached the limit of the rotational operating range of the shaft **2d**.

Modified Examples

[0115] The embodiment disclosed this time must be considered as illustrative in all points and not restrictive. The scope of the present disclosure is not shown by the above description of the embodiment but by the scope of claims for patent, and all modifications or modified examples within the meaning and scope equivalent to the scope of claims for patent are further included.

[0116] While the remote control apparatus **200** includes two operation units **110** to operate two robot arms **50** in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, only one operation unit **110** may alternatively be provided in the remote control apparatus **200** to operate one robot arm **50**.

[0117] While as the controller according to the present disclosure, the first control device **310** arranged in the medical cart **10** is applied in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, a controller other than the first control device **310** may alternatively be applied as the controller according to the present disclosure.

[0118] While the rotational operating range of the shaft **2d** defined on the software is smaller than the mechanical rotational operating range in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, the rotational operating range of the shaft **2d** defined on the software may alternatively be the same as the mechanical rotational operating range.

[0119] While the grip support member **112d** of the operation unit **110** receives an operation to rotate the shaft **2d** in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, an operation to rotate the shaft **2d** may alternatively be received by a structure other than the grip support member **112d**.

[0120] While the drive force of the servomotor **SM7g** acts to prevent rotation of the grip support member **112d** in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, a brake may alternatively be provided on the servomotor **SM7g** to stop rotation of the servomotor **SM7g**.

[0121] While the first control device **310** performs a control to prevent rotation of the links **112a**, **112b**, and **112c** of the wrist **112** when the operation for the instrument **2** received by the operation unit **110** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d** in the aforementioned embodiment, the present disclosure is not limited to this. For example, the first control device **310** may alternatively perform a control to increase resistance to an operation to rotate the links **112a**, **112b**, and **112c** of the wrist **112**. That is, while the links **112a**, **112b**, and **112c** are capable of rotating, resistance to an operation outside the rotational operating range of the shaft **2d** is increased as compared with resistance to an operation within the rotational operating range of the shaft **2d**. Alternatively, it may be possible to select, for each axis of the wrist **112**, a control to prevent the rotation or a control to increase the resistance.

[0122] While an operation on the arm **111** of the operation unit **110** is continuously received even when the operation for the instrument **2** received by the operation unit **110** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d** in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, when the operation for the instrument **2** received by the operation unit **110** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d**, a control may alternatively be performed to prevent rotation of the links **111a**, **111b**, and **111c** of the arm **111**.

[0123] While the operation unit **110** includes seven axes of A21 to A27 in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, the number of axes of the operation unit **110** may alternatively be other than seven. Furthermore, while the wrist **112** of the operation unit **110** includes four axes of A24 to A27 in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, the number of axes of the

wrist **112** may alternatively be other than four, such as three or five.

[0124] While the wrist **112** of the operation unit **110** has a gimbal structure in which the A24 and A26 axes, the A25 axis, and the A27 axis are mutually perpendicular in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, the wrist **112** may not have a gimbal structure. That is, the axes of the wrist **112** may not be mutually perpendicular.

[0125] While operations with the three links **112a**, **112b**, and **112c** are restricted when the operation for the instrument **2** received by the operation unit **110** includes an operation to rotate the shaft **2d** out of the rotational operating range of the shaft **2d** in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, operations with a number of links other than three may alternatively be restricted.

[0126] While the drive forces of the servomotors SM7d, SM7e, and SM7f act to prevent rotation of the links **112a**, **112b**, and **112c** in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, a brake may alternatively be provided on each of the links **112a**, **112b**, and **112c** to stop rotation of links **112a**, **112b**, and **112c**.

[0127] While the message indicating that the operation on the operation unit **110** is an operation out of the rotational operating range of the shaft **2d** is displayed on the monitor **140** of the remote control apparatus **200** and the display **220** arranged on the cart **210** in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, the above message may alternatively be displayed on only one of the monitor **140** and the display **220**.

[0128] While the indicator M indicating the movable range M1 of the robot arm **50**, the movable range M2 of the operation unit **110**, and the current position M3 of the robot arm **50** is displayed on the monitor **140** of the remote control apparatus **200** and the display **220** arranged on the cart **210** in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, the above indicator M may alternatively be displayed on only one of the monitor **140** and the display **220**.

[0129] While four robot arms **50** are provided in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, the number of robot arms **50** may alternatively be any number as long as there is at least one.

[0130] While each of the arm portion **51** and the positioner **30** includes a 7-axis articulated robot in the aforementioned embodiment, the present disclosure is not limited to this. For example, each of the arm portion **51** and the positioner **30** may alternatively include an articulated robot having an axis configuration other than the 7-axis articulated robot. The axis configuration other than the 7-axis articulated robot refers to six axes or eight axes, for example.

[0131] While the surgical robot **100** includes the medical cart **10**, the positioner **30**, the arm base **40**, and the robot arms **50** in the aforementioned embodiment, the present disclosure is not limited to this. For example, the surgical robot **100** may not include the medical cart **10**, the positioner **30**, or the arm base **40**, but may include only the robot arms **50**.

[0132] The functionality of the elements disclosed herein may be implemented using circuitry or processing circuitry that includes general purpose processors, special purpose processors, integrated circuits, application specific integrated circuits (ASICs), conventional circuitry and/or combinations thereof that are configured or programmed to perform the disclosed functionality. Processors are considered processing circuitry or circuitry as they include transistors and other circuitry therein. In the present disclosure, the circuitry, units, or means are hardware that carries out the recited functionality or hardware that is programmed to perform the recited functionality. The hardware may be hardware disclosed herein or other known hardware that is programmed or configured to carry out the recited functionality. When the hardware is a processor that may be considered a type of circuitry, the circuitry, means, or units are a combination of hardware and software, and the software is used to configure the hardware and/or processor.

Aspects

[0133] It will be appreciated by those skilled in the art that the exemplary embodiments described above are specific examples of the following aspects.

(Item 1)

[0134] A robotic surgical system comprising: [0135] a robot arm to allow a surgical instrument including a shaft and a distal end device connected to a distal end of the shaft via a wrist joint to be attached thereto; [0136] an operation apparatus including an operation unit to receive an operation for the surgical instrument; and [0137] a controller configured or programmed to restrict an operation to rotate the shaft in a direction out of a rotational operating range of the shaft with the operation unit and an operation to bend the distal end device with respect to the shaft with the operation unit when the operation for the surgical instrument received by the operation unit includes an operation to rotate the shaft out of the rotational operating range of the shaft.

(Item 2)

[0138] The robotic surgical system according to item 1, wherein the rotational operating range of the shaft is a rotational operating range defined on software that is smaller than a mechanical rotational operating range of the shaft.

(Item 3)

[0139] The robotic surgical system according to item 1 or 2, wherein [0140] the surgical instrument includes a first support member attached to the shaft, and a second support member supported by the first support member so as to be rotatable around a first rotation axis to support the distal end device such that the distal end device is rotatable around a second rotation axis intersecting with the first rotation axis as viewed in a direction in which the shaft extends; and [0141] the operation to bend the distal end device with respect to the shaft includes an operation to rotate the second support member around the first rotation axis and an operation to rotate the distal end device around the second rotation axis.

(Item 4)

[0142] The robotic surgical system according to any one of items 1 to 3, wherein [0143] the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a longitudinal direction of the grip support member to receive an operation to rotate the shaft; and [0144] the operation to rotate the shaft is received by rotating the grip support member around the third rotation axis.

(Item 5)

[0145] The robotic surgical system according to item 4, wherein [0146] the operation unit includes a grip support member motor to rotate the grip support member around the third rotation axis; and [0147] the controller is configured or programmed to control the grip support member motor to prevent rotation of the grip support member when the operation for the surgical instrument includes the operation to rotate the shaft out of the rotational operating range of the shaft.

(Item 6)

[0148] The robotic surgical system according to item 4 or 5, wherein [0149] the operation unit includes an arm and a wrist; [0150] the wrist includes: [0151] a first link including a proximal end connected to a distal end of the arm and rotatable around a fourth rotation axis; [0152] a second link including a proximal end connected to a distal end of the first link and rotatable around a fifth rotation axis; and [0153] a third link including a proximal end connected to a distal end of the second link and a distal end to which the grip support member is connected and rotatable around a sixth rotation axis; and [0154] the controller is configured or programmed to restrict rotation of the first link around the fourth rotation axis, rotation of the second link around the fifth rotation axis, and rotation of the third link around the sixth rotation axis when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft.

(Item 7)

[0155] The robotic surgical system according to item 6, wherein [0156] the wrist includes: [0157] a first link motor to rotate the first link around the fourth rotation axis; [0158] a second link motor to rotate the second link around the fifth rotation axis; and [0159] a third link motor to rotate the third link around the sixth rotation axis; and [0160] the controller is configured or programmed to control the first link motor, the second link motor, and the third link motor to prevent rotation of the first link, the second link, and the third link when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft.

(Item 8)

[0161] The robotic surgical system according to any one of items 1 to 7, comprising: [0162] a first display to display a message indicating that an operation on the operation unit is the operation to rotate the shaft out of the rotational operating range of the shaft when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft.

(Item 9)

[0163] The robotic surgical system according to any one of items 1 to 8, comprising: [0164] a second display to display an indicator indicating a movable range of the robot arm, a movable range of the operation unit with respect to the movable range of the robot arm, and a current position of the robot arm.

(Item 10)

[0165] The robotic surgical system according to any one of items 1 to 9, wherein [0166] the rotational operating range of the shaft is a rotational operating range defined on software that is smaller than a mechanical rotational operating range of the shaft; [0167] the surgical instrument includes a first support member attached to the shaft, and a second support member supported by the first support member so as to be rotatable around a first rotation axis to support the distal end device such that the distal end device is rotatable around a second rotation axis intersecting with the first rotation axis as viewed in a direction in which the shaft extends; and [0168] the operation to bend the distal end device with respect to the shaft includes an operation to rotate the second support member around the first rotation axis and an operation to rotate the distal end device around the second rotation axis.

(Item 11)

[0169] The robotic surgical system according to any one of items 1 to 10, wherein [0170] the rotational operating range of the shaft is a rotational operating range defined on software that is smaller than a mechanical rotational operating range of the shaft; [0171] the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a longitudinal direction of the grip support member to receive an operation to rotate the shaft; and [0172] the operation to rotate the shaft is received by rotating the grip support member around the third rotation axis.

(Item 12)

[0173] The robotic surgical system according to any one of items 1 to 11, wherein [0174] the surgical instrument includes a first support member attached to the shaft, and a second support member supported by the first support member so as to be rotatable around a first rotation axis to support the distal end device such that the distal end device is rotatable around a second rotation axis intersecting with the first rotation axis as viewed in a direction in which the shaft extends; [0175] the operation to bend the distal end device with respect to the shaft includes an operation to rotate the second support member around the first rotation axis and an operation to rotate the distal end device around the second rotation axis; [0176] the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a

longitudinal direction of the grip support member to receive an operation to rotate the shaft; and [0177] the operation to rotate the shaft is received by rotating the grip support member around the third rotation axis.

(Item 13)

[0178] The robotic surgical system according to any one of items 1 to 12, wherein [0179] the rotational operating range of the shaft is a rotational operating range defined on software that is smaller than a mechanical rotational operating range of the shaft; [0180] the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a longitudinal direction of the grip support member to receive an operation to rotate the shaft; [0181] the operation unit includes an arm and a wrist; [0182] the wrist includes: [0183] a first link including a proximal end connected to a distal end of the arm and rotatable around a fourth rotation axis; [0184] a second link including a proximal end connected to a distal end of the first link and rotatable around a fifth rotation axis; and [0185] a third link including a proximal end connected to a distal end of the second link and a distal end to which the grip support member is connected and rotatable around a sixth rotation axis; and [0186] the controller is configured or programmed to restrict rotation of the first link around the fourth rotation axis, rotation of the second link around the fifth rotation axis, and rotation of the third link around the sixth rotation axis when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft.

(Item 14)

[0187] The robotic surgical system according to any one of items 1 to 13, wherein [0188] the surgical instrument includes a first support member attached to the shaft, and a second support member supported by the first support member so as to be rotatable around a first rotation axis to support the distal end device such that the distal end device is rotatable around a second rotation axis intersecting with the first rotation axis as viewed in a direction in which the shaft extends; [0189] the operation to bend the distal end device with respect to the shaft includes an operation to rotate the second support member around the first rotation axis and an operation to rotate the distal end device around the second rotation axis; [0190] the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a longitudinal direction of the grip support member to receive an operation to rotate the shaft; [0191] the operation unit includes an arm and a wrist; [0192] the wrist includes: [0193] a first link including a proximal end connected to a distal end of the arm and rotatable around a fourth rotation axis; [0194] a second link including a proximal end connected to a distal end of the first link and rotatable around a fifth rotation axis; and [0195] a third link including a proximal end connected to a distal end of the second link and a distal end to which the grip support member is connected and rotatable around a sixth rotation axis; and [0196] the controller is configured or programmed to restrict rotation of the first link around the fourth rotation axis, rotation of the second link around the fifth rotation axis, and rotation of the third link around the sixth rotation axis when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft.

(Item 15)

[0197] The robotic surgical system according to any one of items 1 to 14, wherein [0198] the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a longitudinal direction of the grip support member to receive an operation to rotate the shaft; [0199] the operation to rotate the shaft is received by rotating the grip support member around the third rotation axis; [0200] the operation unit includes an arm and a wrist; [0201] the wrist includes: [0202] a first link including a proximal end connected to a distal end of

the arm and rotatable around a fourth rotation axis; [0203] a second link including a proximal end connected to a distal end of the first link and rotatable around a fifth rotation axis; and [0204] a third link including a proximal end connected to a distal end of the second link and a distal end to which the grip support member is connected and rotatable around a sixth rotation axis; and [0205] the controller is configured or programmed to restrict rotation of the first link around the fourth rotation axis, rotation of the second link around the fifth rotation axis, and rotation of the third link around the sixth rotation axis when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft.

(Item 16)

[0206] The robotic surgical system according to any one of items 1 to 15, comprising: [0207] a first display to display a message indicating that an operation on the operation unit is the operation to rotate the shaft out of the rotational operating range of the shaft when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft; and [0208] a second display to display an indicator indicating a movable range of the robot arm, a movable range of the operation unit with respect to the movable range of the robot arm, and a current position of the robot arm.

(Item 17)

[0209] The robotic surgical system according to item 16, wherein [0210] the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a longitudinal direction of the grip support member to receive an operation to rotate the shaft; and [0211] the operation to rotate the shaft is received by rotating the grip support member around the third rotation axis.

(Item 18)

[0212] The robotic surgical system according to item 16 or 17, wherein [0213] the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a longitudinal direction of the grip support member to receive an operation to rotate the shaft; [0214] the operation unit includes an arm and a wrist; [0215] the wrist includes: [0216] a first link including a proximal end connected to a distal end of the arm and rotatable around a fourth rotation axis; [0217] a second link including a proximal end connected to a distal end of the first link and rotatable around a fifth rotation axis; and [0218] a third link including a proximal end connected to a distal end of the second link and a distal end to which the grip support member is connected and rotatable around a sixth rotation axis; and [0219] the controller is configured or programmed to restrict rotation of the first link around the fourth rotation axis, rotation of the second link around the fifth rotation axis, and rotation of the third link around the sixth rotation axis when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft.

(Item 19)

[0220] A control method for a robotic surgical system, the robotic surgical system comprising a robot arm to allow a surgical instrument including a shaft and a distal end device connected to a distal end of the shaft via a wrist joint to be attached thereto, and an operation apparatus including an operation unit to receive an operation for the surgical instrument, the control method comprising: [0221] receiving the operation for the surgical instrument by the operation unit; and [0222] restricting an operation to rotate the shaft in a direction out of a rotational operating range of the shaft with the operation unit and an operation to bend the distal end device with respect to the shaft with the operation unit when the operation for the surgical instrument received by the operation unit includes an operation to rotate the shaft out of the rotational operating range of the shaft.

(Item 20)

[0223] A storage medium operable to store a program for a control method for a robotic surgical system, the robotic surgical system comprising a robot arm to allow a surgical instrument including a shaft and a distal end device connected to a distal end of the shaft via a wrist joint to be attached thereto, and an operation apparatus including an operation unit to receive an operation for the surgical instrument, the storage medium operable to store the program for the control method, the control method comprising: [0224] receiving the operation for the surgical instrument by the operation unit; and [0225] restricting an operation to rotate the shaft in a direction out of a rotational operating range of the shaft with the operation unit and an operation to bend the distal end device with respect to the shaft with the operation unit when the operation for the surgical instrument received by the operation unit includes an operation to rotate the shaft out of the rotational operating range of the shaft.

Claims

1. A robotic surgical system comprising: a robot arm to allow a surgical instrument including a shaft and a distal end device connected to a distal end of the shaft via a wrist joint to be attached thereto; an operation apparatus including an operation unit to receive an operation for the surgical instrument; and a controller configured or programmed to restrict an operation to rotate the shaft in a direction out of a rotational operating range of the shaft with the operation unit and an operation to bend the distal end device with respect to the shaft with the operation unit when the operation for the surgical instrument received by the operation unit includes an operation to rotate the shaft out of the rotational operating range of the shaft.
2. The robotic surgical system according to claim 1, wherein the rotational operating range of the shaft is a rotational operating range defined on software that is smaller than a mechanical rotational operating range of the shaft.
3. The robotic surgical system according to claim 1, wherein the surgical instrument includes a first support member attached to the shaft, and a second support member supported by the first support member so as to be rotatable around a first rotation axis to support the distal end device such that the distal end device is rotatable around a second rotation axis intersecting with the first rotation axis as viewed in a direction in which the shaft extends; and the operation to bend the distal end device with respect to the shaft includes an operation to rotate the second support member around the first rotation axis and an operation to rotate the distal end device around the second rotation axis.
4. The robotic surgical system according to claim 1, wherein the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a longitudinal direction of the grip support member to receive an operation to rotate the shaft; and the operation to rotate the shaft is received by rotating the grip support member around the third rotation axis.
5. The robotic surgical system according to claim 4, wherein the operation unit includes a grip support member motor to rotate the grip support member around the third rotation axis; and the controller is configured or programmed to control the grip support member motor to prevent rotation of the grip support member when the operation for the surgical instrument includes the operation to rotate the shaft out of the rotational operating range of the shaft.
6. The robotic surgical system according to claim 4, wherein the operation unit includes an arm and a wrist; the wrist includes: a first link including a proximal end connected to a distal end of the arm and rotatable around a fourth rotation axis; a second link including a proximal end connected to a distal end of the first link and rotatable around a fifth rotation axis; and a third link including a proximal end connected to a distal end of the second link and a distal end to which the grip support

member is connected and rotatable around a sixth rotation axis; and the controller is configured or programmed to restrict rotation of the first link around the fourth rotation axis, rotation of the second link around the fifth rotation axis, and rotation of the third link around the sixth rotation axis when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft.

7. The robotic surgical system according to claim 6, wherein the wrist includes: a first link motor to rotate the first link around the fourth rotation axis; a second link motor to rotate the second link around the fifth rotation axis; and a third link motor to rotate the third link around the sixth rotation axis; and the controller is configured or programmed to control the first link motor, the second link motor, and the third link motor to prevent rotation of the first link, the second link, and the third link when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft.

8. The robotic surgical system according to claim 1, comprising: a first display to display a message indicating that an operation on the operation unit is the operation to rotate the shaft out of the rotational operating range of the shaft when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft.

9. The robotic surgical system according to claim 1, comprising: a second display to display an indicator indicating a movable range of the robot arm, a movable range of the operation unit with respect to the movable range of the robot arm, and a current position of the robot arm.

10. The robotic surgical system according to claim 1, wherein the rotational operating range of the shaft is a rotational operating range defined on software that is smaller than a mechanical rotational operating range of the shaft; the surgical instrument includes a first support member attached to the shaft, and a second support member supported by the first support member so as to be rotatable around a first rotation axis to support the distal end device such that the distal end device is rotatable around a second rotation axis intersecting with the first rotation axis as viewed in a direction in which the shaft extends; and the operation to bend the distal end device with respect to the shaft includes an operation to rotate the second support member around the first rotation axis and an operation to rotate the distal end device around the second rotation axis.

11. The robotic surgical system according to claim 1, wherein the rotational operating range of the shaft is a rotational operating range defined on software that is smaller than a mechanical rotational operating range of the shaft; the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a longitudinal direction of the grip support member to receive an operation to rotate the shaft; and the operation to rotate the shaft is received by rotating the grip support member around the third rotation axis.

12. The robotic surgical system according to claim 1, wherein the surgical instrument includes a first support member attached to the shaft, and a second support member supported by the first support member so as to be rotatable around a first rotation axis to support the distal end device such that the distal end device is rotatable around a second rotation axis intersecting with the first rotation axis as viewed in a direction in which the shaft extends; the operation to bend the distal end device with respect to the shaft includes an operation to rotate the second support member around the first rotation axis and an operation to rotate the distal end device around the second rotation axis; the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a longitudinal direction of the grip support member to receive an operation to rotate the shaft; and the operation to rotate the shaft is received by rotating the grip support member around the third rotation axis.

13. The robotic surgical system according to claim 1, wherein the rotational operating range of the shaft is a rotational operating range defined on software that is smaller than a mechanical rotational

operating range of the shaft; the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a longitudinal direction of the grip support member to receive an operation to rotate the shaft; the operation unit includes an arm and a wrist; the wrist includes: a first link including a proximal end connected to a distal end of the arm and rotatable around a fourth rotation axis; a second link including a proximal end connected to a distal end of the first link and rotatable around a fifth rotation axis; and a third link including a proximal end connected to a distal end of the second link and a distal end to which the grip support member is connected and rotatable around a sixth rotation axis; and the controller is configured or programmed to restrict rotation of the first link around the fourth rotation axis, rotation of the second link around the fifth rotation axis, and rotation of the third link around the sixth rotation axis when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft.

14. The robotic surgical system according to claim 1, wherein the surgical instrument includes a first support member attached to the shaft, and a second support member supported by the first support member so as to be rotatable around a first rotation axis to support the distal end device such that the distal end device is rotatable around a second rotation axis intersecting with the first rotation axis as viewed in a direction in which the shaft extends; the operation to bend the distal end device with respect to the shaft includes an operation to rotate the second support member around the first rotation axis and an operation to rotate the distal end device around the second rotation axis; the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a longitudinal direction of the grip support member to receive an operation to rotate the shaft; the operation unit includes an arm and a wrist; the wrist includes: a first link including a proximal end connected to a distal end of the arm and rotatable around a fourth rotation axis; a second link including a proximal end connected to a distal end of the first link and rotatable around a fifth rotation axis; and a third link including a proximal end connected to a distal end of the second link and a distal end to which the grip support member is connected and rotatable around a sixth rotation axis; and the controller is configured or programmed to restrict rotation of the first link around the fourth rotation axis, rotation of the second link around the fifth rotation axis, and rotation of the third link around the sixth rotation axis when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft.

15. The robotic surgical system according to claim 1, wherein the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a longitudinal direction of the grip support member to receive an operation to rotate the shaft; the operation to rotate the shaft is received by rotating the grip support member around the third rotation axis; the operation unit includes an arm and a wrist; the wrist includes: a first link including a proximal end connected to a distal end of the arm and rotatable around a fourth rotation axis; a second link including a proximal end connected to a distal end of the first link and rotatable around a fifth rotation axis; and a third link including a proximal end connected to a distal end of the second link and a distal end to which the grip support member is connected and rotatable around a sixth rotation axis; and the controller is configured or programmed to restrict rotation of the first link around the fourth rotation axis, rotation of the second link around the fifth rotation axis, and rotation of the third link around the sixth rotation axis when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft.

16. The robotic surgical system according to claim 1, comprising: a first display to display a message indicating that an operation on the operation unit is the operation to rotate the shaft out of

the rotational operating range of the shaft when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft; and a second display to display an indicator indicating a movable range of the robot arm, a movable range of the operation unit with respect to the movable range of the robot arm, and a current position of the robot arm.

17. The robotic surgical system according to claim 16, wherein the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a longitudinal direction of the grip support member to receive an operation to rotate the shaft; and the operation to rotate the shaft is received by rotating the grip support member around the third rotation axis.

18. The robotic surgical system according to claim 16, wherein the operation unit includes a grip support member to rotatably support one end of a grip member to be operated by a finger of an operator, the grip support member being operable to rotate around a third rotation axis parallel to a longitudinal direction of the grip support member to receive an operation to rotate the shaft; the operation unit includes an arm and a wrist; the wrist includes: a first link including a proximal end connected to a distal end of the arm and rotatable around a fourth rotation axis; a second link including a proximal end connected to a distal end of the first link and rotatable around a fifth rotation axis; and a third link including a proximal end connected to a distal end of the second link and a distal end to which the grip support member is connected and rotatable around a sixth rotation axis; and the controller is configured or programmed to restrict rotation of the first link around the fourth rotation axis, rotation of the second link around the fifth rotation axis, and rotation of the third link around the sixth rotation axis when the operation for the surgical instrument received by the operation unit includes the operation to rotate the shaft out of the rotational operating range of the shaft.

19. A control method for a robotic surgical system, the robotic surgical system comprising a robot arm to allow a surgical instrument including a shaft and a distal end device connected to a distal end of the shaft via a wrist joint to be attached thereto, and an operation apparatus including an operation unit to receive an operation for the surgical instrument, the control method comprising: receiving the operation for the surgical instrument by the operation unit; and restricting an operation to rotate the shaft in a direction out of a rotational operating range of the shaft with the operation unit and an operation to bend the distal end device with respect to the shaft with the operation unit when the operation for the surgical instrument received by the operation unit includes an operation to rotate the shaft out of the rotational operating range of the shaft.

20. A storage medium operable to store a program for a control method for a robotic surgical system, the robotic surgical system comprising a robot arm to allow a surgical instrument including a shaft and a distal end device connected to a distal end of the shaft via a wrist joint to be attached thereto, and an operation apparatus including an operation unit to receive an operation for the surgical instrument, the storage medium operable to store the program for the control method, the control method comprising: receiving the operation for the surgical instrument by the operation unit; and restricting an operation to rotate the shaft in a direction out of a rotational operating range of the shaft with the operation unit and an operation to bend the distal end device with respect to the shaft with the operation unit when the operation for the surgical instrument received by the operation unit includes an operation to rotate the shaft out of the rotational operating range of the shaft.
